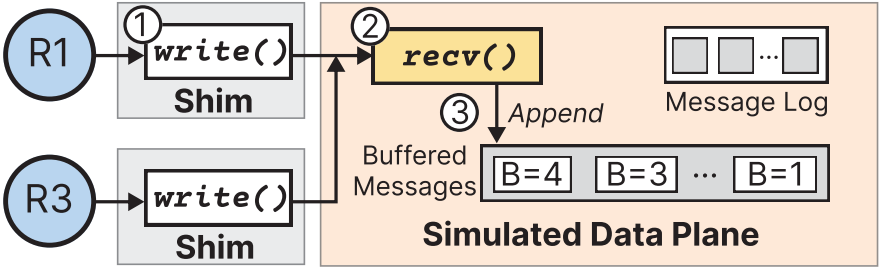
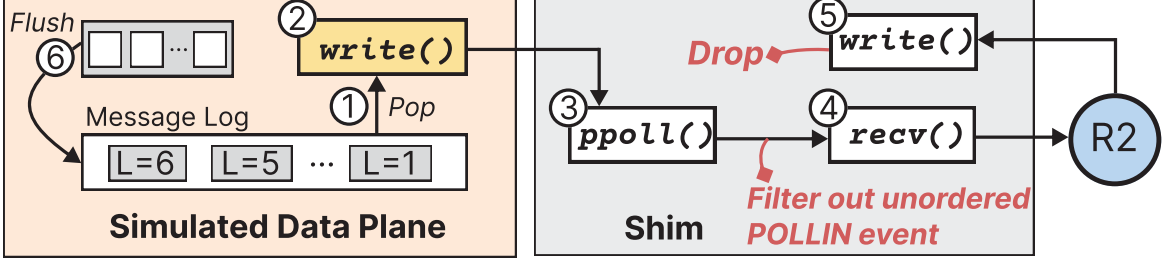


(a) Snapshot *R2* by recording its received messages.



(b) Keeping connection and buffering messages while *R2* is offline.



(c) Restoring *R2* by replaying the recorded messages.